

DETOUR

Shannon's Theorem (1948)

*What does it mean for a model to be
"surprised" by the data it sees?*

The answer is rooted in the foundations of information theory.

The Three Axioms

Shannon asked: is there a unique way to measure the uncertainty of an event that satisfies certain axiomatic properties?

Property	Formal Statement	Geometric Meaning
Continuity	Small $\Delta p \rightarrow$ small $\Delta I(p)$	Curve has no jumps or breaks
Monotonicity	$p < q \rightarrow I(p) > I(q)$	Slopes down; steeper toward zero; no flat regions, no reversals
Additivity	$I(p \cdot q) = I(p) + I(q)$	Independent events $\rightarrow$ additive info; rules out all functions except log

Result: the unique function satisfying all three is $I(p) = -\log(p)$

Self-Information and Entropy

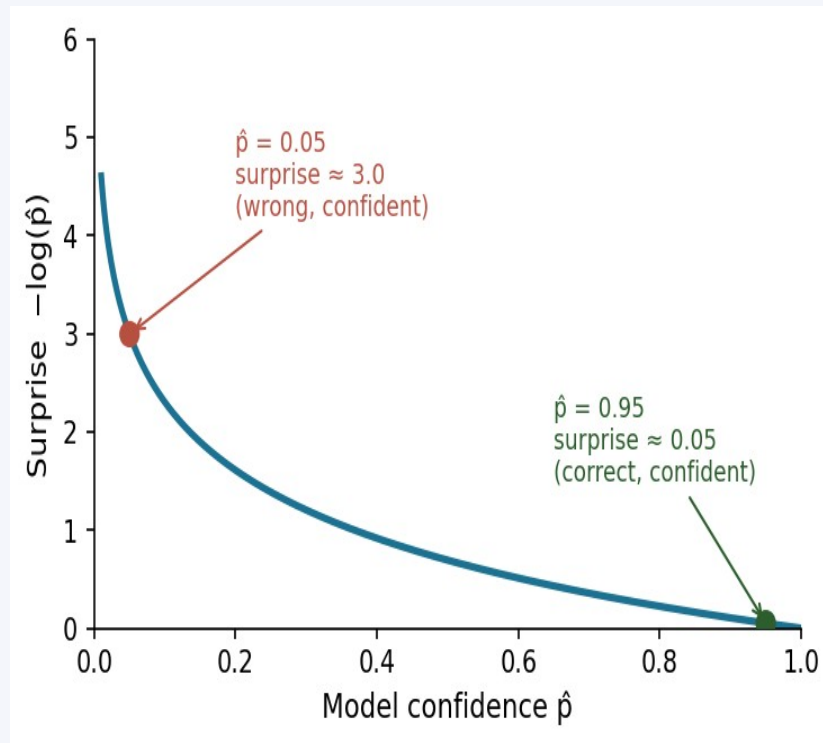
Self-information $I(p(x)) = -\log p(x)$

- ▶ Surprise of observing one specific outcome
- ▶ Property of a single event — measured after it occurs
- ▶ Confident & correct $\rightarrow$ low surprise (≈ 0)
- ▶ Confident & wrong $\rightarrow$ high surprise ($\rightarrow \infty$)

Entropy $H = -\sum p(x) \log p(x)$

- ▶ Expected self-information over the whole distribution
- ▶ Average surprise before any outcome is observed
- ▶ Property of the distribution — not of any single event

Shannon's theorem: H is the unique consistent measure of distributional uncertainty.



FROM SHANNON TO ML

The Model's Surprise

Why does a model trained on imbalanced data fail to recognize the minority class — even when the loss appears low?

To answer this precisely, we need to understand what the loss function is actually measuring.

Cross-Entropy: Measuring the Model's Surprise

True distribution $p(x)$

$p(x) = 1$ for the correct class
 $p(x) = 0$ for all others (one-hot spike)

Model belief $q(x)$

Probability the model assigns to each outcome based on learned weights

$$H(p, q) = -\sum p(x) \log q(x)$$

$p(x)$ weights which outcomes actually matter (the true labels)
 $\log q(x)$ measures how surprised the model is by each one

Training = minimizing average surprise across all m examples:

$$J = -(1/m) \sum_i \log q(y^{(i)} | x^{(i)})$$

$q(y^{(i)} | x^{(i)})$ = probability the model assigned to the correct label y , given input x

Class Imbalance Corrupts the Average

Majority class (e.g. 'no fraud')

- ▶ Model quickly learns to be confident
- ▶ Predicted probability $q(x)$ is high
- ▶ Surprise $-\log q(x)$ is near zero
- ▶ Contributes small loss — for 95% of samples

Minority class (e.g. 'fraud')

- ▶ Model assigns low probability $q(x)$
- ▶ Surprise $-\log q(x)$ is very large
- ▶ But contributes only 5% of the sum
- ▶ High surprise is diluted — nearly invisible

Low J does not mean a good model — it may mean a model that has learned to ignore the minority class entirely.